3AL Tutorial 11 2024

1. Use the Third Sylow Theorem to prove that a group of order 77 is cyclic.

2. Let G be a group of order p^2q for distinct primes p and q with $p < q$. Prove that if G has more than one Sylow q-subgroup, then $|G| = 12$.

3. Let G be a group of order p^2q^2 for distinct primes p and q with $q \nmid (p^2 - 1)$ and $p \nmid (q^2 - 1)$.

   (a) Let P and Q be normal subgroups of G with orders p^2 and q^2, respectively.

      (i) Show that $P \cap Q = \{1\}$.

      (ii) Prove that $G = PQ$.

      (iii) Prove that for $x \in P$ and $y \in Q$, we have $xy = yx$.

   (b) Prove that G is abelian.

   HINT: First find subgroups P and Q that satisfy the conditions of (a).

4. Let G be a group of order $70 = 2 \times 5 \times 7$. Show that G has a normal subgroup of index 2.

   HINT: Find subgroups P and Q such that $|G : PQ| = 2$.

5. Prove that there is no simple group of order $825 = 3 \times 5^2 \times 11$.

6. Let R be a ring.

   (a) Prove that the identity of R is unique.

   (b) Prove that if an element $r \in R$ has a multiplicative inverse, then it is unique.

   (c) Prove that $(-1_R)(-1_R) = 1_R$.

7. Let R be a ring and $n \geq 2$. Suppose that $r^n = r$ for all $r \in R$. Let $a, b \in R$. Prove that $ab = 0$ implies $ba = 0$.

8. The **characteristic** of a ring R is the smallest *positive* integer n such that $n1_R = 0_R$. If such a positive integer does not exist, the characteristic of R is said to be 0. The characteristic of R is denoted by $\mathrm{char}(R)$.

(a) Let $m \geq 2$. Determine $\text{char}(\mathbb{Z}_m)$.

(b) Let R be a ring with $\text{char}(R) = n$. What can we say about $\text{char}(M_2(R))$?

(c) Prove that if $n = \text{char}(R)$ then $nr = 0_R$ for all $r \in R$.

Extra problems

(I) Let G be a group with order $p^n m$ where $n \geq 1$ and p is a prime such that $p \nmid m$. Suppose that H is a normal subgroup of G such that p divides the order of H. Prove that every Sylow p-subgroup of G/H can be written as PH/H for some Sylow p-subgroup P of G.

HINT: Think about orders, Correspondence Theorem, Lagrange's Theorem, First Sylow Theorem, Second Isomorphism Theorem.